

PET final-design selection study

Summary (generated from committed results)

PET final-design study

PET is diagnostic method development. Nothing here is a publication adoption; no historical threshold, verdict or the predecessor campaign's disposition changes; simulation only.

What the evidence supports

question	what the evidence supports
Can improved PET recover meaningful physics changes? Is it more accurate than current GBDT?	Yes, in this simulation test. H2S1T24 $K=5$ passed every frozen point-estimator rule: energy tilt 0.895, proton topology 0.391, energy $\times q_3$ 0.446. Suggestive, not decisive. On the development tilt (DEV bank) scalar GBDT OmniFold reached 0.783 ($k=3$) to 0.817 ($k=10$), H2S1T24 0.904; inputs, k and events are not matched. Efficiency-corrected binned IBU reached 0.923 on the tilt but fails the proton case.
Can it give tighter, reliable confidence intervals? Will it distinguish generators more strongly?	Unestablished. Its six-member bootstrap interval failed calibration: too wide overall (68 % intervals cover 0.896), too narrow at low acceptance (0.257). Unestablished. That needs calibrated inference including detector and model uncertainties; no tested procedure provides it.

Terminal outcome by the frozen rules: **NO_ELIGIBLE_DESIGN**, no selected default. Signal-only simulation, conditional on the frozen event banks; PET is diagnostic.

How the numbers are measured

- ▶ **Simulation only.** Pseudodata are simulated events with a known, injected change to the truth (e.g. an E_{avail} tilt, more protons); the unfolding starts from the unchanged simulation (the prior).
- ▶ **Recovery** $R = 1 - L_1(\text{unfolded} - \text{target}) / L_1(\text{prior} - \text{target})$ on normalized histograms: 1 = the injected change fully recovered, 0 = no better than the prior, < 0 = moved away.
- ▶ **Tests:** energy tilt and its opposite (E0, E3), proton topology (E4, $E_{\text{avail}} \times$ proton class), energy $\times$ momentum transfer (E5), a 21-case robustness library, weight tails and seed spread.
- ▶ **Fresh events:** designs were developed on a DEV bank; the final test used a bank no earlier work had scored, blinded until every final run was complete. A third bank stays sealed.
- ▶ **Rules frozen first:** every threshold, finalist and sample size was committed before final results were seen; the outcome is whatever those rules give.
- ▶ **Coverage:** does the six-member bootstrap interval contain the true value at its nominal rate across 120 independent replicates?

Design names

name part	meaning
CTL	historical production PET recipe: misses carried, 3 iterations, 8 epochs, forced low rate after iteration 0
C	CTL with the predecessor's efficiency-corrected truth step
B	predecessor's feature arm: reco energy summaries at step 1, one-hot particle type at step 2
H1	C + categorical truth particle type (13-class one-hot) in the truth step
H2	H1 + reco summaries at the detector step (reco E_{avail} , reco q_3 , cluster ΣE and count)
S1	constant learning rate at every iteration
E16 / T24	16 epochs per fit at both steps / 24 epochs per truth-step fit
L64 / L128	wider detector-step PET (projection 64 / 128, 4 layers)
P2pre / P2scr	Gregor's PET2-small backbone at the detector step on its 33-token typed inputs, pretrained / from scratch
A1	annealed detector-step learning rate after iteration 1
X4	4 independently seeded truth-step fits averaged per iteration
$K=5$, $K=4$	the frozen number of OmniFold iterations

Names compose: $H2S1T24 = H2 + S1 + T24$. Step 1 reweights simulation to match (pseudo)data at detector level; step 2 turns those weights into truth-level weights; one pass is one iteration.

Decision rules (frozen in advance)

1. **Eligibility first.** A design must pass every rule at its own frozen settings:
 - *accuracy*: recovery floors on the energy tilt, its opposite, the regions, proton topology and energy $\times q_3$, at a simultaneous 95 % lower bound;
 - *robustness*: across a 21-case library it rarely moves away from the truth;
 - *stability*: no extreme weights, and a repeat run with new seeds agrees to within 0.05;
 - *calibration*: its uncertainty intervals cover the truth at their nominal rate.
2. **Then choose.** The smaller network wins only if it is as accurate (within 0.02) and at least twice as cheap; otherwise the stronger, larger design.
3. **No eligible design means no winner** — and no default.

Every threshold, finalist and sample size was frozen before any final-bank result was seen.

Architectures tested, what came from Gregor's work, and why none passed

design	change	from Gregor's work	why it did not pass
H1	C + categorical truth PDG (13-way one-hot)	type encoding (cmp 2), padding kept distinct; truth step is ours	S-U4 fails at every k (best 0.023; need ≥ 0.25)
H2	H1 + reco summaries at step 1	whole-event energy sums + overflow (cmp 4, 6; not per-family)	no k passes all four screens
CS1	C + constant rate	—	S-N2 seed sd 0.054 > 0.05
H2S1	H2 + constant rate	as H2	S-N2 seed sd 0.095 > 0.05
H2S1E16	H2S1, 16 epochs	as H2	S-N2 seed sd 0.074 > 0.05
L64H2	H2, step-1 PET width 64	as H2 + one capacity step (cmp 7)	no k passes all four screens
L128H2	H2, step-1 PET width 128	as H2 + one capacity step (cmp 7)	no k passes all four screens
L64S1	H2S1, width 64	as L64H2	S-N2 seed sd 0.057 > 0.05
L128S1	H2S1, width 128	as L128H2	S-N2 seed sd 0.091 > 0.05
L128S1E16	L128S1, 16 epochs	as L128H2	S-N2 seed sd 0.071 > 0.05
P2preS1	PET2-small at step 1, pretrained	his backbone, checkpoint, 33-token typed inputs, recipe	S-N1 step-1 weight tail at $K^* = 4$
P2scrS1	PET2-small, random init	as P2preS1 without the checkpoint	S-U4 fails at every k (best 0.110; need ≥ 0.25)
P2preA1	P2preS1, annealed step-1 rate	as P2preS1	S-U4 fails at every k (best 0.225; need ≥ 0.25)
P2scrA1	P2scrS1, annealed step-1 rate	as P2scrS1	S-U4 fails at every k (best 0.060; need ≥ 0.25)
H2S1T24	H2S1, 24-epoch truth step (finalist)	as H2	final bank: coverage C1, C4 FAIL — INELIGIBLE (C1, C4)
L128S1T24	L128S1, 24-epoch truth step (finalist)	as L128H2	final bank: B2 FAIL (point-decided) — INELIGIBLE (B2)

Also closed: the step-2 ensemble fallback (H2S1X4, L128S1X4; entry condition never met) and AUSSIE (bounded matched scalar test). Anchors/controls (CTL, C, B) are not selectable. Development screens: S-U1 tilt, S-U3 opposite tilt, S-U4 proton topology, S-B2 D4d residual; then S-N1, S-N2. Gregor-derived parts follow the pinned comparison (cmp sections); the truth-step design is the study's own.

The two finalists side by side

quantity (final bank, look 1)	H2S1T24 $K=5$	L128S1T24 $K=4$
Energy-tilt recovery (E0)	0.895	0.863
Proton-topology recovery (E4)	0.391	0.348
Joint energy-momentum-transfer recovery (E5)	0.446	0.450
Estimator-seed spread of energy recovery (N2)	0.037	0.043
Largest observed truth weight (N1)	90.2	33.4
Measured A100-hours per unfolding	1.77	1.52
Frozen point-estimator rules	all pass	B2 fails
Interval calibration (coverage)	fails (C1, C4)	not tested (ineligible on B2)

Recoveries: 1 = exact, 0 = no better than the prior. B2's failure is a point decision whose interval crosses the limit; the two designs are not shown to differ in robustness.

Uncertainty calibration: coverage of H2S1T24

H2S1T24 $K=5$, $B = 6$, 120 replicates	measured	verdict
C1 pooled 95 % (LB ≥ 0.90 , point ≤ 0.99)	0.990 (LB 0.980)	unresolved
C1 pooled 68 % (LB ≥ 0.60 , point ≤ 0.80)	0.896 (LB 0.858)	decisive
C1 (both levels)		FAIL
C2 every bin 95 % ≥ 0.85	all bins pass	PASS
C3 moderate / good 95 % LB ≥ 0.85	0.949 / 0.993	PASS
C4 half-width / limit, bins 1–6 (≤ 1)	1.32–1.72	FAIL
C5 D4c up	skipped after decisive C1–C4 FAIL	—

In aggregate the six-member interval is **too wide**; the same against the FB population target. Not uniformly conservative: the low-acceptance region **under-covers** (68 %: 0.26, 95 %: 0.79), a region no rule gates. Mechanism not established.

Reference methods and alternatives

method	what it is	from Gregor's work	result
CTL $K=3$	historical production PET recipe (anchor)	—	final bank energy tilt 0.306 (floor 0.556)
C $K=3$	efficiency-corrected truth step (anchor)	—	final bank energy tilt 0.560
B	predecessor's feature arm (control)	energy sums (cmp 4)	development tilt 0.474 at $k=3$
Gregor's arm	his PET2-small, his tokens, pretrained; historical comparison, $k=3$	his configuration (adapted)	0.417 vs ours 0.304; floor 0.556: NEITHER ELIGIBLE
GBDT OmniFold	scalar HistGradientBoosting on reco p_T , $p_{ }$, E_{avail} , q_3 , cluster sums	—	development tilt 0.783 (efficiency-corrected) / 0.418 (misses carried), $k=3$; proton topology 0.389 only with species-count truth inputs
binned IBU	response-matrix unfolding, efficiency-corrected	—	development tilt 0.923; moves away in the proton case (-0.381)
AUSSIE	scalar method of arXiv:2602.24282, matched to GBDT OmniFold	—	lost all 12 matched comparisons on robustness and stability

Anchors are not selectable; scalar references use the predecessor's DEV-bank selections, so they are context, not matched comparisons with the PET finalists.

Appendix

Backup

Algorithmic alternative: AUSSIE (scalar, matched)

AUSSIE setting vs OmniFold $k=3$	moves-away: AUSSIE	OmniFold	worst R : AUSSIE	OmniFold
$\lambda=1000$, misses carried, kinematics	11	6	-1.077	-0.217
$\lambda=1000$, misses carried, kinematics + species	8	6	-0.985	-0.230
$\lambda=0$, efficiency-corrected, kinematics	14	6	-3.876	-0.392
$\lambda=0$, efficiency-corrected, kinematics + species	7	6	-0.735	-0.352

Bounded matched stress test (same scalar inputs, miss handling, data and tuning opportunity); truth kinematics = $(E_{\text{avail}}, p_T, p_{\parallel}, q_3)$. 12 comparisons: 2 miss rules $\times$ 2 truth-input sets $\times$ 3 OmniFold operating points (shown: $k=3$). AUSSIE wins none of the 12 on robustness or stability, so it does not advance to a PET backbone.